Nokia BLR-CORE-02 Core Node Hardware/Environmental Fault Resolution SOP

Node: BLR-CORE-02 | **Vendor:** Nokia | **Severity:** Warning | **Alarm Code:** ALARM_4539

Description

The failure of Node ID: BLR-CORE-02, a Nokia Core node at an unknown site, due to Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss) and alarm code ALARM_4539 with a severity of Warning, is a complex issue. This occurs due to the intricate protocol mechanisms and hardware components of the Nokia Core node, specifically the Line Card (LC) and Power Supply (PS) redundancy. The Line Card is responsible for processing and forwarding data packets, while the Power Supply redundancy ensures uninterrupted power to the node. However, in the event of a hardware failure, such as ASIC memory parity errors, or environmental issues, such as power supply redundancy loss, the node may experience a Line Card crash, leading to a loss of connectivity and triggering the ALARM_4539 alarm code.

Pre-requisite Checks

- Verify the site location and ensure remote hands are dispatched to the unknown site.
- Check the redundant link status to ensure that it is operational and providing connectivity.
- Perform a visual inspection of the node to identify any physical damage or issues.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Error	High
Line Card Crash	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show card detail, Expected Output: Card 0/1/1: Line Card, Online, Description: This command provides detailed information about the Line Card, including its status and operational state. It will help identify if the Line Card is online and functioning correctly.

Command: show power status, Expected Output: Power Supply 0: Online, Redundancy: Active/Active, Description: This command provides information about the Power Supply status, including its operational state and redundancy configuration. It will help identify if the Power Supply is online and functioning correctly.

Command: show alarm history, Expected Output: Alarm Code: ALARM_4539, Time: 2023-02-20 14:30:00, Description: This command provides a history of alarm codes, including ALARM_4539, and the time at which
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they occurred. It will help identify the timing and frequency of the
alarm.
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Resolution Steps

Step 1: Step: Power Cycle the Node, Command: reboot node force, Description: This step involves forcefully rebooting the node to reset its operational state and reload its configuration. It will help resolve any transient issues and restore connectivity.

Step 2: Step: Replace the Faulty Line Card, Command: replace card 0/1/1, Description: This step involves replacing the faulty Line Card with a new one to restore processing and forwarding capabilities. It will help resolve any hardware-related issues.

Step 3: Step: Restore Power Supply Redundancy, Command: configure power supply redundancy active/active, Description: This step involves restoring the Power Supply redundancy to ensure uninterrupted power to the node. It will help resolve any power-related issues.

Post-Check Verification

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Command: show card detail, Expected Output: Card 0/1/1: Line Card,
Online, Description: This command provides detailed information about
the Line Card, including its status and operational state. It will
help verify that the Line Card is online and functioning correctly.

Command: show power status, Expected Output: Power Supply 0: Online,
Redundancy: Active/Active, Description: This command provides
information about the Power Supply status, including its operational
state and redundancy configuration. It will help verify that the Power
Supply is online and functioning correctly.

Command: show alarm history, Expected Output: Alarm Code: ALARM_4539,
Time: 2023-02-20 14:30:00, Description: This command provides a
history of alarm codes, including ALARM_4539, and the time at which
they occurred. It will help verify that the alarm has been cleared.
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Roll-back Procedures

- Step: Revert to Previous Configuration, Command: configure revert, Description: This step involves reverting to the previous configuration to restore the node to its previous operational state. It will help resolve any issues caused by the fix.

- Step: Reboot the Node, Command: reboot node, Description: This step involves rebooting the node to reload its configuration and restore its operational state. It will help resolve any issues caused by the fix.